

ORDYNSKOYE, Russian Federation

WMO: 297260

Lat: 54.3653N

Lon: 81.8842E

Elev: 132

StdP: 99.74

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.3	-33.8	-40.2	0.1	-37.1	-36.7	0.1	-33.7	7.5	-9.3	6.9	-9.0	0.3	60	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	29.7	18.6	28.2	18.2	26.6	17.5	20.4	26.4	19.5	25.5	18.7	24.6	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.5	23.0	17.4	12.7	22.3	16.5	11.9	21.5	59.5	26.4	56.5	25.4	54.0	24.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.8	5.2	DB	-40.0	33.0	3.8	2.1	-42.7	34.5	-45.0	35.8	-47.1	37.0	-49.9	38.5	(4)
(5)			WB	-40.0	21.6	3.6	1.1	-42.6	22.4	-44.7	23.0	-46.7	23.7	-49.3	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-18.2	-15.1	-6.6	3.9	11.9	17.8	19.5	16.9	10.5	3.5	-6.4	-14.0	(6)	
	DBStd	14.48	8.97	7.81	6.76	5.40	5.26	4.19	3.18	3.62	4.42	4.90	7.91	8.49	(7)	
	HDD10.0	3822	874	702	513	193	43	2	0	1	46	208	493	745	(8)	
	HDD18.3	6076	1132	936	772	433	206	59	24	70	236	461	743	1003	(9)	
	CDD10.0	926	0	0	0	11	102	236	294	216	62	6	0	0	(10)	
	CDD18.3	139	0	0	0	0	7	42	61	27	2	0	0	0	(11)	
	CDH23.3	1369	0	0	0	5	136	418	486	281	44	0	0	0	(12)	
	CDH26.7	330	0	0	0	1	32	101	118	69	9	0	0	0	(13)	
(14) Wind	WSAvg	2.2	2.1	2.1	2.3	2.4	2.4	2.0	1.8	1.8	2.0	2.3	2.5	2.5	(14)	
(15) Precipitation	PrecAvg	431	21	17	19	23	36	52	73	49	41	35	34	31	(15)	
	PrecMax	587	37	34	36	51	82	95	146	121	97	84	69	63	(16)	
	PrecMin	303	4	1	4	3	5	15	20	6	9	13	18	11	(17)	
	PrecStd	80	11	8	9	12	20	25	34	27	21	18	14	13	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.3	2.0	9.6	23.5	30.1	31.1	32.2	31.8	28.0	19.7	9.5	2.0	(19)	
		MCWB	0.2	0.0	4.6	12.0	16.0	18.5	19.7	19.6	16.2	11.5	6.5	0.7	(20)	
	2%	DB	-1.2	0.0	6.2	19.4	26.7	29.3	29.4	28.7	23.9	16.3	7.1	0.4	(21)	
		MCWB	-2.3	-1.3	2.7	10.6	14.6	17.9	19.6	18.8	14.9	9.8	4.5	-0.5	(22)	
	5%	DB	-3.0	-1.6	4.1	16.0	24.4	27.6	27.7	26.5	20.9	13.4	4.7	-1.1	(23)	
		MCWB	-3.9	-2.7	1.4	9.3	13.8	17.7	19.0	17.8	13.5	8.5	2.9	-2.0	(24)	
	10%	DB	-5.1	-3.6	2.3	13.1	21.7	25.9	26.1	24.4	18.5	10.6	2.7	-2.9	(25)	
		MCWB	-6.0	-4.6	0.4	7.6	12.8	17.1	18.4	16.9	12.5	7.2	1.2	-3.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	0.7	5.1	13.5	17.4	20.2	22.0	21.1	17.1	12.2	7.1	0.9	(27)	
		MCDB	1.2	1.6	9.2	21.1	26.8	27.1	28.2	27.5	23.7	18.0	8.8	1.5	(28)	
	2%	WB	-2.0	-1.2	3.0	11.3	15.9	19.4	20.9	19.7	15.3	10.4	4.6	-0.3	(29)	
		MCDB	-1.3	-0.1	5.4	17.7	23.8	26.2	26.6	26.5	22.4	15.2	6.5	0.4	(30)	
	5%	WB	-3.9	-2.7	1.7	9.5	14.6	18.6	20.0	18.6	14.1	8.9	3.1	-2.0	(31)	
		MCDB	-2.9	-1.6	3.8	15.6	22.4	25.4	25.5	24.9	20.0	12.7	4.8	-1.2	(32)	
	10%	WB	-5.9	-4.6	0.5	7.8	13.5	17.8	19.2	17.6	12.8	7.4	1.3	-3.8	(33)	
		MCDB	-5.1	-3.6	2.2	12.9	20.8	24.2	24.6	23.4	17.5	10.6	2.6	-2.9	(34)	
(35) Mean Daily Temperature Range	MDBR		9.4	10.6	10.7	10.5	12.1	10.7	9.8	10.8	10.4	8.2	6.6	8.0	(35)	
	5% DB	MCDBR	8.2	9.0	11.3	16.2	16.6	13.9	12.3	14.4	15.4	14.2	7.1	7.7	(36)	
		MCWBR	7.7	8.1	8.3	8.9	7.1	5.3	4.6	6.1	7.6	8.4	5.3	7.4	(37)	
	5% WB	MCDBR	8.2	8.6	9.8	15.1	13.9	11.6	10.1	12.1	13.4	12.7	6.6	7.2	(38)	
		MCWBR	7.8	7.9	7.5	8.7	6.6	4.9	4.5	5.5	7.0	8.0	5.3	7.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.296	0.329	0.380	0.410	0.410	0.411	0.437	0.413	0.372	0.355	0.331	0.298	(40)	
	taud		2.011	1.961	1.904	2.097	2.203	2.287	2.213	2.255	2.317	2.235	2.053	1.960	(41)	
	Ebn at Noon		603	715	763	809	835	838	805	801	787	697	550	494	(42)	
	Edh at Noon		74	110	147	140	133	124	131	118	98	84	71	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.01	2.06	3.52	4.71	5.51	6.10	5.47	4.67	3.21	1.63	0.80	0.66	(44)	
	RadStd		0.08	0.12	0.27	0.28	0.59	0.52	0.35	0.33	0.35	0.23	0.13	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page